INHERITED FROM RADICALS

Rational exponents inherit the domain restrictions of radicals. Whether a negative base is allowed depends on the denominator of the exponent.

RATIONAL EXPONENT

$$a^{m/n}$$

split by **whether the denominator n is even or odd**

≥0

CASE 1

n even

Base must be non-negative

no real number squares to a negative number; the expression would require complex numbers

E.G. $(-4)^{1/2}$ is not real

$\mathbb{R}$

CASE 2

n odd

Any real base

negative bases have real odd roots

E.G. $(-8)^{1/3} = -2$

ONE MORE RESTRICTION

When the exponent is negative, $a \neq 0$ is required as well: division by zero is undefined. The restrictions apply equally to $\sqrt[n]{a}$ and $a^{1/n}$.